# Clinical Term Mapping Engine: ICD-10-CM & SNOMED CT

HiLabs Hackathon 2026 · technical write-up · demo video: (link to be added) · repo: dhisanthks/hilabs (private)

All numbers are 5-fold cross-validated on the training data. The evaluation set contains **diagnosis / condition terms only**, so every SNOMED CT number below is on diagnosis terms (all ICD rows are diagnoses). **Headline (the lower of the two systems, as the brief scores it): 67.4%, set by SNOMED CT** — which is why most late work targeted SNOMED. SNOMED partial credit uses the semantic tag because the supplied dictionary has no IS-A relationships, so hierarchy distance cannot be computed without a UMLS/SNOMED licence.

| 5-fold CV (diagnosis terms) | ICD-10-CM | SNOMED CT |
|---|---|---|
| **Exact match** | **71.1%** | **67.4%** |
| Codes never seen in training (long tail) | 50.1% | 57.4% |
| Rare or never-seen codes (≤ 2 training examples) | 61.5% | 63.3% |
| Partial credit: ICD 3-character category / SNOMED semantic tag | 81.6% | 84.4% |
| Exact or near-synonym (name similarity ≥ 90) | 74.4% | 73.9% |

## 1. Approach — problem and what the data told us

Input: an entity extracted by CodEx plus its source chunk; output: one ICD-10-CM code and one SNOMED CT concept from the supplied dictionaries (98,222 ICD codes; 382,124 SNOMED concepts, 984,585 descriptions). 9,646 labelled rows (2,306 ICD, 7,340 SNOMED of which 2,700 diagnosis terms) shaped every decision:

• **Every entity string occurs once** — lookup is useless; the engine must generalize to unseen wording.

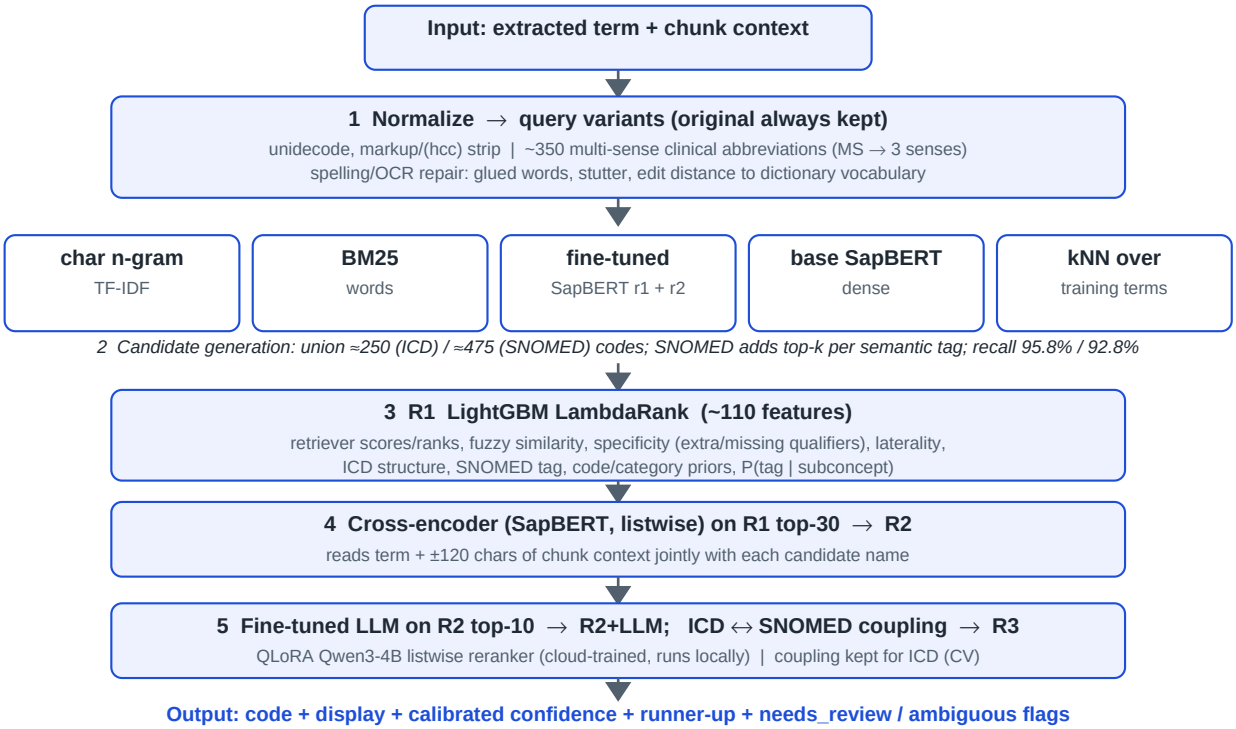
• **Local conventions dominate** — only 6% (ICD) / 15% (SNOMED) of terms match a dictionary string exactly (*can not squat* → R26.2 Difficulty in walking; *rashes* → Eruption).

• **Long tail** — 24% (ICD) / 46% (SNOMED diagnosis) of held-out terms have a gold code never seen in training folds.

• **Label ambiguity** — see §6: the gold standard's own consistency is 93.7% / 84.6%.

**Protocol.** 5-fold CV grouped by source chunk; every learned component (encoders, cross-encoder, LLM, kNN memory, priors, rankers) is re-trained inside each fold, so every number is on terms the system never saw.

## 2. Architecture




## 3. How accuracy was built up (each step measured, 5-fold CV)

| Step | ICD | SNOMED (diagnosis) |
|---|---|---|
| char TF-IDF + nearest-training-term baseline | 41.2 | n/m* |
| + normalization + R1 LambdaRank over lexical retrievers | 55.9 | 53.4 |
| + SapBERT dense retrieval and kNN (best of 7 encoders benchmarked) | 63.1 | n/m* |
| + per-fold fine-tuned SapBERT + tag-stratified SNOMED retrieval | 66.7 | 62.0 |
| + cross-encoder with chunk context (R2) | 67.5 | 62.3 |
| + ICD↔SNOMED coupling (R3) | 67.7 | 62.2 |
| + round-2 fine-tuning (iterative hard negatives), ensembled with round-1 encoder | 68.3 | 62.7 |
| + cross-encoder v2 (negatives from the final ranker, 2 epochs) | 68.4 | 63.2 |

| Step | ICD | SNOMED (diagnosis) |
|---|---|---|
| + chunk-context features (rankers see the chart context) | 68.9 | 63.6 |
| **+ fine-tuned LLM reranker (R2+LLM) — final** | **71.1** | **67.4** |

* n/m: these two early stages were scored only on all SNOMED rows, not on the diagnosis subset.

## 4. How the models are trained

• **Fine-tuned SapBERT bi-encoders (retrieval + kNN features, feed R1).** Start from SapBERT (PubMedBERT trained on UMLS synonyms). Contrastive fine-tuning on (term → gold code description) pairs with MultipleNegativesRanking loss: every other pair in the batch is a negative, plus explicit hard negatives; 2 epochs, lr 2e-5, batch 64, re-trained per fold. *Round 1* uses the lexically closest wrong code as the hard negative. *Round 2* uses the 3 wrong codes the round-1 model itself ranked highest (iterative hard negatives) — typically sibling codes differing only in specificity (E11.40 vs E11.42, unspecified vs specific), so the encoder learns the coders' specificity and wording conventions. Both rounds are kept as complementary feature sets (ensemble).

• **R1 / R2 / R2+LLM / R3 — LightGBM LambdaRank.** Each term's candidates form one ranking group (label 1 = gold). LambdaRank grows trees that push the gold above wrong candidates, weighting each pair by its effect on top-1 (NDCG). This is how retriever weights are learned: the 'weight' of a retriever is not a fixed number — trees split on each retriever's score and rank *in combination* with other features (e.g. trust nearest-training-term votes when several neighbours agree; distrust dense similarity when the candidate adds an unsupported qualifier), and the criterion is purely which combination puts the gold first on training folds. A fixed quality-weighted fusion was tested and lost (38.0% vs 51.5% top-1). Each ranker is 5-seed bagged (70% rows / features per model).

• **Cross-encoder (R2).** SapBERT as a pair classifier reading [term | ±120 characters of chunk context] with each candidate name; listwise softmax loss over the gold + 9 hard negatives taken from R1's own top-30; 2 epochs, per fold.

• **Fine-tuned LLM reranker (R2+LLM).** Qwen3-4B-Instruct-2507 (Apache-2.0) loaded in 4-bit and fine-tuned with QLoRA (LoRA rank 16, α 32, on all attention and MLP projections; ~0.8% of weights trainable). Prompt: coding system, entity type, term, chart context and R2's top-10 candidates lettered A–J; the training loss is cross-entropy over the 10 letter logits at the answer position (listwise, no text generation); candidate order is shuffled during training so it cannot learn position. 1 epoch, lr 1e-4, honest 5-fold on Kaggle 2× T4 + a final adapter trained on all data. At inference one forward pass gives a log-probability per candidate (deterministic) → features llm_lp / llm_rank / llm_gap for the R2+LLM ranker. Gain: +2.2 ICD, +3.8 SNOMED diagnosis.

## 5. Design decisions (and the stage each one lives in)

• **[Retrieval] Many cheap retrievers, union of candidates** (char n-gram, BM25, fine-tuned + base SapBERT, kNN over training terms; SNOMED also top-k inside each semantic tag) → 96% of gold codes in the pool.

• **[Normalization] Variants, never replacements** — abbreviation senses (MS → 3 senses), spelling/OCR repair, glued words.

• **[R1] Retriever weighting learned, not hand-set** — LambdaRank over ~110 per-candidate features (see §4).

• **[R1] Specificity calibration** — handled by the round-2 fine-tuned SapBERT (trained against sibling codes) plus explicit R1 features: qualifier words a candidate adds (over-coding) or misses (under-coding), unspecified / other / leaf flags. Specificity errors are balanced (46 over- vs 60 under-coded), i.e. no systematic bias.

• **[R1] Context** — chunk-context type classifier and chunk-style flags let the ranker use the chart context.

• **[R2] Cross-encoder** reads term + context jointly with each candidate — resolves abbreviation collisions from context.

• **[R2+LLM] LLM where judgement is needed** — only on R2's top-10, as features, not as a hard decision.

• **[R3] Independent prediction, then coupling** — a learned crosswalk recovers only 37% of ICD codes even from the gold SNOMED code, so each system is predicted on its own; the other system's top-5 enters R3 as evidence. Kept for ICD, not for SNOMED (final stage R2+LLM).

• **[Deployment] Deterministic, private, batch-oriented** — fixed seeds, no sampling; one command; everything runs locally on a 4 GB laptop GPU (RTX 3050). Latency, both systems per term: one term at a time median 3.7 s / p95 4.4 s with the LLM (2.7 / 3.0 s without); in batch mode, as a chart pipeline runs, ~0.85 s/term with the LLM (~0.2 s without) because dictionary search and GPU work are shared across the batch — ~45 min per 3,000 terms on one laptop GPU, and files can be split across GPUs. Dictionary files are fingerprinted for version drift.

## 6. Error analysis, label ambiguity and flagging

| Failure mode (share of errors) | ICD (667) | SNOMED diagnosis (879) |
|---|---|---|
| gold never retrieved | 14% | 13% |
| right ICD category, wrong specificity | 34% | — |
| right chapter, wrong category / same SNOMED tag, wrong concept | 24% | 48% |
| wrong chapter / wrong SNOMED semantic tag (mostly disorder ↔ finding) | 28% | 39% |
| …of which the predicted name is a near-synonym of the gold (≥ 90) | 10% | 20% |

• **Label ambiguity.** Near-identical training terms share the majority gold code only **93.7% (ICD) / 84.6% (SNOMED)** of the time (*ble edema* → R60.9 vs *mild ble edema* → R60.0; *phq-9* vs *phq 9* → different concepts) — matching the ~0.85 inter-annotator agreement of the SNOMED CT Entity Linking Challenge. 20% of SNOMED diagnosis errors are near-synonyms (*Hyperhidrosis* vs *Diaphoresis*); crediting them gives 73.9%. Twin-preference features and label cleaning could not learn the coders' tie-breaking (±0.2).

• **Hard slices** (Appendix B): ICD chapters Z 47.5% / T 51.4%; SNOMED gold types situation 18.5% / procedure 20.6%; context abbreviations on ICD (*MS* in an echo report → multiple sclerosis).

• **Flagging mechanism.** confidence = softmax probability of the chosen code over the final ranker's scores. *needs_review* = confidence < 0.5 → flags 30% of SNOMED diagnosis terms, which are only 32.8% correct, while the unflagged 70% are 82.4% correct. *ambiguous* =

runner-up probability ≥ half the winner's, or runner-up name a near-synonym (≥ 90); every row also carries the runner-up code (alt_code). Confidence > 0.9 is right 96.1% (SNOMED dx).

• **Coverage / abstention.** The engine never leaves a code blank — the headline scores every term and even flagged terms are right 32.8% of the time, so abstaining would lower it (70% coverage × 82.4% = 57.7% < 67.4%). Instead the confidence column lets a user abstain at any threshold, e.g. send the least-confident 10% or 30% to coders:

| Terms attempted (most confident first) | 100% | 90% | 70% | 50% | 30% |
|---|---|---|---|---|---|
| SNOMED CT (diagnosis) — accuracy on attempted terms | 67.4% | 72.4% | 82.4% | 90.1% | 95.9% |
| ICD-10-CM — accuracy on attempted terms | 71.1% | 76.3% | 85.4% | 92.3% | 96.4% |

## 7. Core requirement: the seven hard cases

| Failure mode | How the engine handles it | Evidence |
|---|---|---|
| 1 Nearest string is wrong | ~110 signals beyond string similarity; specificity features penalize extra qualifiers | HTN → I10; pulmonary hypertension → I27.20 |
| 2 Abbreviation collision | all senses retrieved; cross-encoder + LLM read the chunk context; low confidence → review | MS (echo) → mitral valve stenosis |
| 3 Specificity calibration | round-2 fine-tuned SapBERT (sibling negatives) + R1 add/miss-qualifier features | diabetes → E11.9; t2dm with diabetic neuropathy → E11.40 |
| 4 Granularity mismatch | both systems predicted independently; coupling only as R3 evidence | crosswalk alone: 37% |
| 5 Long tail | dictionary-wide retrieval, fine-tuned encoders, LLM knowledge | never-seen codes 50.1% / 57.4% |
| 6 Upstream noise | markup strip, glued-word split, stutter cut, spelling repair (as variants) | chronickidney disease stag 3 → N18.30 |
| 7 Version drift | dictionary fingerprint → all indexes rebuilt; new codes retrievable at once | README §Version drift |

## 8. Answers to the brief's open questions

• **Partial credit?** Exact match as headline; report hierarchical credit (ICD category 81.6%, SNOMED tag 84.4%).

• **Ambiguous gold?** Accept sets of valid codes (20% of SNOMED diagnosis errors are near-synonyms); rows carry alt_code.

• **Inter-annotator agreement?** Publish it as the ceiling: 93.7% / 84.6% measured; 95% exact match exceeds it.

• **Per-category floors?** Most ICD chapters 67–82%; a ~50% floor would flag chapter Z (47.5%) and rare SNOMED gold types.

## 9. What we tried and abandoned

| Idea | Result | Decision |
|---|---|---|
| Zero-shot local LLM judge (Qwen3-4B, few-shot) | CV blend +0.25 | replaced by fine-tuned LLM |
| 2nd LLM: SNOMED-diagnosis specialist over the top-20 | stacked 67.44% = without it | dropped (redundant) |
| Cross-encoder v3 with the entity type in the query | ICD +0.6, SNOMED –0.7 | dropped |
| Quality-weighted retriever fusion (RRF, MRR weights) | 38.0% vs 51.5% top-1; as feature ±0.2 | dropped |
| Concept-type classifier / two-level type ranker | type accuracy = R1's own | dropped |
| Context-kNN; twin / majority-convention features; cleaned labels | ±0.2 (noise) | dropped |
| Diagnosis-only specialized ranker | 62.9% vs 63.2% (all-rows ranker) | dropped |

## 10. Challenges & learnings

• **The data sets the ceiling** — measure label consistency first. **New signals beat new features** — fine-tuning, cross-encoder and LLM gave the gains; eight feature ideas gave ≈ 0. **Trust only full 5-fold numbers** — one fold overstated the cross-encoder (+3.5 vs +0.6).

• **A 4 GB GPU shaped the engineering:** two disk-full incidents (page-file growth), memory-mapped caches, atomic saves, two-phase inference (5.7 → 0.85 s/term).

## 11. Innovations

Tag-stratified SNOMED retrieval; iterative hard-negative fine-tuning (two-round encoder ensemble); explicit specificity features; a listwise QLoRA-tuned LLM reranker read by letter log-probabilities (deterministic, runs locally); calibrated review/ambiguity flags; measured label-consistency ceiling; dictionary fingerprinting.

## 12. Future improvements — a plan for the remaining gap, by error bucket

| SNOMED dx errors (879) | Next step | Why it should help |
|---|---|---|
| Wrong semantic tag, 39% (disorder ↔ finding) | UMLS/SNOMED licence → IS-A hierarchy; twin-aware features, hierarchical scoring | often the same clinical idea under two tags |
| Same tag, wrong concept, 48% | stronger LLM judge (7–8B, more epochs) applied only to the 30% flagged terms | the 4B LLM gave the largest late gain (+3.8); the gold is in the top-10 for 88% of terms |
| Gold never retrieved, 13% | synthetic chart-style training terms for unseen codes (local LLM, no patient data); more retrievers | 46% of terms have a gold code never seen in training |
| Label ambiguity (ceiling 84.6%) | set-based gold; published conventions; coder feedback on flagged terms | raises the measurable ceiling; feedback becomes training data |

## 13. External resources

SapBERT (cambridgeltl/SapBERT-from-PubMedBERT-fulltext, Apache-2.0) — bi-encoders and cross-encoder. Qwen3-4B-Instruct-2507 (Apache-2.0; unsloth 4-bit build) fine-tuned with QLoRA (PEFT, bitsandbytes) on Kaggle 2× T4 GPUs. Benchmarked only: BioLORD-2023, gte-base, MedCPT, BGE-base-en-v1.5, MedEmbed-base-v0.1, e5-base-v2. ICD-10-CM and SNOMED CT US dictionaries as supplied; hand-curated clinical abbreviation list; PyTorch, sentence-transformers, transformers, LightGBM, scikit-learn, RapidFuzz. Related work: DrivenData SNOMED CT Entity Linking Challenge, n2c2 2019 Track 3, xMEN, MedCAT (docs/RESEARCH.md).

**Reproduce:** see README — `python run.py predict --input eval.csv --output predictions.csv --assume-subconcept DiagnosisCode` (weights pulled automatically from huggingface.co/Dhisanth/hilabs-term-mapping); **accuracy:** `python scripts/score.py --pred outputs/predictions.csv --gold gold.csv` (exact match per system + headline).

## Appendix A — Long-tail generalization (accuracy by gold-code frequency in the training folds)

| Gold code seen in training | ICD share | ICD accuracy | SNOMED dx share | SNOMED dx accuracy |
|---|---|---|---|---|
| absent (never) | 23.8% | 50.1% | 46.2% | 57.4% |
| rare (1–2 times) | 24.4% | 72.6% | 29.8% | 72.4% |
| 3–5 times | 21.9% | 80.0% | 16.4% | 80.8% |
| 6–20 times | 29.2% | 80.0% | 7.6% | 80.9% |
| rare or absent (≤ 2) | 48% | 61.5% | 76% | 63.3% |

## Appendix B — Accuracy by category (final model)

| ICD chapter | n | accuracy | SNOMED diagnosis gold type | n | accuracy |
|---|---|---|---|---|---|
| N / E / B | 68 / 131 / 30 | 82.4 / 80.2 / 80.0% | disorder | 1,276 | 76.1% |
| C / G / K | 50 / 79 / 102 | 78.0 / 77.2 / 76.5% | finding | 1,035 | 68.5% |
| J / I / R | 143 / 292 / 690 | 74.1 / 74.0 / 72.0% | situation | 65 | 18.5% |
| M / D / H / F | 293 / 28 / 62 / 46 | 67.9 / 67.9 / 67.7 / 67.4% | procedure | 34 | 20.6% |
| S / L / T / Z | 50 / 86 / 37 / 80 | 66.0 / 62.8 / 51.4 / 47.5% | | | |

## Appendix C — Zero-shot encoder benchmark (retrieval recall over all rows, kNN@1)

| Encoder | ICD R@1 | ICD R@100 | ICD kNN@1 | SNOMED R@1 | SNOMED R@100 | SNOMED kNN@1 |
|---|---|---|---|---|---|---|
| SapBERT (chosen) | 32.8 | 76.5 | 48.5 | 38.6 | 79.2 | 26.7 |
| BioLORD-2023 | 33.6 | 81.1 | 46.5 | 36.7 | 75.4 | 25.3 |
| gte-base | 28.3 | 75.3 | 47.1 | 38.0 | 80.2 | 25.8 |
| MedCPT | 25.3 | 79.3 | 47.2 | 35.2 | 77.6 | 25.0 |